Project Title: Who Gets to Do Nothing? A Comparative Analysis (Team #37)
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1. Introduction

Leisure is an important part of someone's day, because it brings about a range of physical, psychological and social benefits. However, different groups of society face large disparities in how much leisure they enjoy. We want to dig deeper to better understand these disparities in leisure time for two reasons - one, so that individuals themselves can understand where they fare relative to different segments of society, and two, so that healthcare professionals are better equipped to target leisure outreach initiatives in society.

2. Problem Statement

Jargon-free Problem: We aim to understand who in the United States has less time to relax each day and why. Using American Time Use Survey (ATUS) data, we will examine how demographic and socioeconomic factors (eg. age, gender) relate to daily leisure time, in order to identify populations with limited leisure time and reveal daily schedule patterns that shape access to relaxation.

Formal Problem: We aim to study how people's characteristics relate to their daily leisure time. Using ATUS data, we define the outcome to be daily leisure time, and the inputs to be individual-level factors like age, gender, income, employment status and education. We model leisure time as a function of these factors using a supervised machine learning model (e.g. regression). We also cluster individuals based on time-use proportions to uncover common daily schedule types. We will compare leisure time across clusters, test for significant differences, and analyze demographic composition. Success will be evaluated through model performance, cluster interpretability, and usability of the visual analytics interface.

3. Literature Survey

Existing research demonstrates how leisure and relaxation play a key role in individual well-being. Using structural equation modeling, [1] finds that engagement in leisure activities strengthens resilience and reduces the likelihood of psychological problems. [2] reviews empirical evidence connecting leisure activities to improved physical and cognitive health, including reduced risk of age-related diseases. [3] develops a framework explaining how leisure activities induce pleasant moods and emotions, contributing to overall health. [4] shows that relaxation engagement can foster psychological states such as core mindfulness and positive energy, while [5] links relaxation to reductions in anxiety, cognitive burden, and tension headaches. [6] demonstrates that time pressure is associated with depression and helps explain gender differences in psychological distress, suggesting that access to discretionary time connects daily schedules to mental health. More broadly, [7] introduces the National Time Accounting framework, emphasizing that leisure and relaxation are central to societal welfare. While this body of work provides compelling motivation for focusing on relaxation time, it does not address how access to leisure and its benefits vary systematically across population groups.

A related body of literature explores disparities in discretionary time across demographic and socioeconomic groups. [8] introduces the concept of time poverty as a complement to official poverty metrics, suggesting that household burdens are not captured by financial measures alone. Subsequent empirical studies using time-use data show how time poverty varies across demographic groups. Using Canadian survey data, [9] estimates time-adjusted poverty rates and finds substantial time deficits among

employed single parents. [10] conducts a logistic regression analysis using UK Time Use Survey data to show that working women experience greater constraints on their free time. Using Ghana Living Standards Survey data, [11] finds that gender, residence, education, and income are key determinants of time poverty, while [12] employs probit analysis on Korea Labor and Income Panel data to show that gender and marital status affect time poverty across income groups. Similarly, [13] reveals persistent inequalities in schedule intensity across gender and occupations using UK time-use data. This literature provides a foundation for understanding disparities in time availability but does not focus specifically on relaxation time or the full range of factors affecting it.

Studies using ATUS data typically combine time-allocation theory with econometric methods to estimate average time allocation patterns. [14] standardizes activity codes across time-use surveys and applies regression and decomposition techniques to estimate long-run trends in leisure across demographic groups. [15] jointly models leisure participation and duration through a multinomial probit and duration regression, showing that separately estimating participation and time spent on leisure can produce biased results. More recent ATUS-based work such as [16] links social determinants such as education, poverty, and food insecurity to variation in time spent socializing, relaxing, and engaging in leisure using survey-weighted regression models. Related work such as [17] argues that simple distinctions between work and leisure are insufficient for understanding well-being and proposes alternative ways of grouping activities based on how people feel while spending time. Building on this concept, [18] applies machine learning methods to ATUS data, using conditional random forests to predict whether a day is rated "better than typical," identifying diminishing marginal returns in time allocation for activities such as work and socialization. Although these approaches identify average demographic effects and long-run shifts in leisure allocation, they primarily estimate mean relationships rather than discovering latent behavioral archetypes emerging from the full structure of daily schedules.

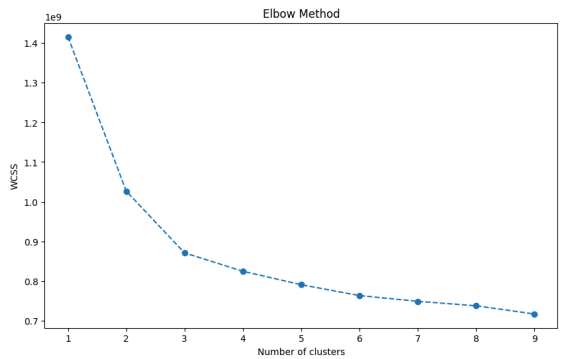
4. Proposed Method

Our proposed computational and visualization methods are novel as it relates to time-use data analysis. In this section, we will describe our methods in detail and then list the innovations in our approach.

Our computation methods can be divided into two parts. First, we do clustering on individuals' daily time schedules (specifically on the amount of time they spend on each activity in the ATUS dataset). We do not include any demographic variables as variables in the clustering algorithm because we want to uncover latent behavioral archetypes by grouping individuals with similar time structures together. We implemented a K-means clustering algorithm and used an elbow diagram to determine the optimal number of clusters, which turned out to be **3 clusters**. We then examined each cluster, using statistical testing, to identify the features that most significantly differentiated clusters from one another.

Second, we use a supervised machine learning model to predict leisure time for an individual based on a number of relevant demographic and socioeconomic features. We do this to better understand which features most strongly correlate with leisure time and which features are less important. Through feature importance and regression results, we can quantify the extent to which each variable contributes to the model's predictions. We first did an 80-20 train-test split of the data. The baseline OLS model explained 23.45% of the variation in daily leisure time (test $R^2 = 0.2345$), with all coefficients statistically significant ($p < 0.05$). The improved model with more regressors increased explanatory power to 25.20%

(test $R^2 = 0.2520$), with nearly all coefficients remaining statistically significant. We tried Random Forest, Quantile Regression, and Lasso, but chose the improved OLS model as it yielded the best result.



OLS Regression Results — Improved Model

Coefficient Estimates

Variable	Coef.	Std. Err.	t-stat	P>|t|
const	**183.2*** **	(4.821)	38.00	<0.001
TEAGE_C	**1.421*** **	(0.052)	27.33	<0.001
TEAGE_C_SQ	**0.041*** **	(0.004)	10.25	<0.001
FEMALE	**-61.6*** **	(1.201)	-51.29	<0.001
TRCHILDNUM	**-17.4*** **	(0.876)	-19.86	<0.001
TRHOLIDAY	**22.1*** **	(3.512)	6.29	<0.001
EARNINGS	**-0.021*** **	(0.003)	-7.00	<0.001
EARNINGS_MISSING	**18.3*** **	(2.143)	8.54	<0.001
PART_TIME	**31.2*** **	(1.654)	18.86	<0.001
MULTI_JOB	**-34.3*** **	(2.011)	-17.06	<0.001
FEMALE_X_CHILDNUM	**-2.2*** **	(0.743)	-2.96	0.003
AGE_X_EMPLOYMENT	0.084	(0.881)	0.10	0.912
CHILD_X_NOPARTNER	**-3.2*** **	(1.102)	-2.90	0.004
TELFS_2	**60.1*** **	(4.213)	14.26	<0.001
TELFS_3	**104.3*** **	(9.021)	11.56	<0.001
TELFS_4	**110.2*** **	(3.154)	34.94	<0.001
TELFS_5	**143.1*** **	(1.421)	100.70	<0.001
TUDIARYDAY_2	**-54.3*** **	(2.031)	-26.74	<0.001
TUDIARYDAY_3	**-63.9*** **	(2.037)	-31.37	<0.001
TUDIARYDAY_4	**-64.8*** **	(2.028)	-31.95	<0.001
TUDIARYDAY_5	**-69.1*** **	(2.043)	-33.82	<0.001
TUDIARYDAY_6	**-55.4*** **	(2.051)	-27.01	<0.001
TUDIARYDAY_7	**12.8*** **	(2.055)	6.23	<0.001
TRSPPRES_2	**38.4*** **	(2.934)	13.09	<0.001
TRSPPRES_3	**42.8*** **	(1.291)	33.15	<0.001
RACE_2	**33.1*** **	(1.823)	18.16	<0.001
RACE_3	**-18.2*** **	(2.541)	-7.16	<0.001
RACE_4	7.1	(4.712)	1.51	0.088
EDUC_2	**-8.4*** **	(1.654)	-5.08	<0.001
EDUC_3	**-18.3*** **	(1.721)	-10.63	<0.001
EDUC_4	**-31.2*** **	(1.843)	-16.93	<0.001
EDUC_5	**-46.1*** **	(2.011)	-22.92	<0.001
YEAR_2011	-1.2	(2.341)	-0.51	0.610
YEAR_2012	-0.8	(2.298)	-0.35	0.727
YEAR_2013	1.1	(2.312)	0.48	0.634
YEAR_2014	0.3	(2.287)	0.13	0.896
YEAR_2015	2.1	(2.301)	0.91	0.361
YEAR_2016	1.8	(2.315)	0.78	0.436
YEAR_2017	2.4	(2.298)	1.04	0.297
YEAR_2018	4.1*	(2.287)	1.79	0.073
YEAR_2019	5.2**	(2.301)	2.26	0.024
YEAR_2021	**24.3*** **	(2.512)	9.67	<0.001
YEAR_2022	**15.1*** **	(2.487)	6.07	<0.001
YEAR_2023	**13.2*** **	(2.498)	5.28	<0.001

**** p < 0.001 ** p < 0.01 * p < 0.05 (HC3 robust standard errors)*

Model Summary & Diagnostics

OLS Improved Model | Dependent variable: TOTAL_LEISURE (min/day)

Fit Statistics	
N	**116,198**
R^2	**0.252**
Adj. R^2	**0.252**
F-statistic	**893.4**** **
Prob > F	< 0.001
AIC	1,537,000
BIC	1,538,000
Df Model	44

Residual Diagnostics	
Omnibus	**14,201*** **
Prob (Omnibus)	< 0.001
JB Test	**22,841*** **
Prob (JB)	< 0.001
Skew	0.87
Kurtosis	4.31
Durbin-Watson	2.00
Cond. Number	412

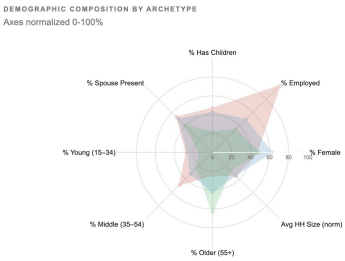
**** p < 0.001 **** p < 0.0001*

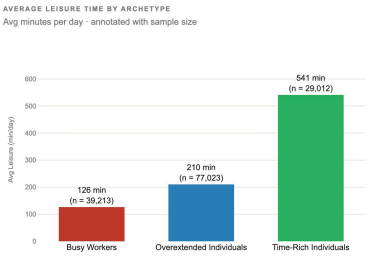
<u>Novel Ideas in Computation Methods:</u> (1) Clustering on time-use information to uncover latent behavioral archetypes; (2) Statistical testing to uncover significant differences between each cluster.

Our visualization methods can similarly be split into two parts. To visualize the clustering results, we developed three visualizations. First, a bar chart displays the average daily leisure time (in minutes) for each cluster, where bar size encodes the leisure time amount and each bar is annotated with its sample size so viewers understand how many respondents each cluster is based on. Second, a radar chart shows the demographic composition of each cluster across eight axes: % Female, % Employed, % Has Children, % Spouse Present, % Young (15–34), % Middle (35–54), % Older (55+), and average household size. Together, these visualizations allow users to see both how much leisure time each group has and what kinds of people make up each cluster.

To visualize our regression results, we constructed sampling distributions of the predicted mean leisure time using the OLS-estimated coefficients and standard errors. By the Central Limit Theorem, the sampling distribution of the estimated group mean is approximately normal for sufficiently large samples, regardless of the underlying distribution of individual leisure times. We display two curves: a fixed baseline curve representing the OLS reference profile (male, employed at work, spouse present, no children, Monday), and a dynamic curve that updates based on the demographic filters the user selects. This allows users to directly compare the predicted leisure time for any demographic subgroup against the baseline profile, and also the uncertainty around them and the spread of plausible predicted values.

<u>Novel Ideas in Visualization Methods:</u> (1) Radar charts to visualize demographic differences among the clusters; (2) Normal curves instead of regression coefficients to capture uncertainty in OLS estimates.





5. Evaluation

For our clustering method, we used internal validity indices to evaluate our clusters in the absence of ground truth labels. In particular, we obtained a Silhouette Score of 0.2375 and a Davies-Bouldin Index of 1.5475. These results indicate moderate cluster separation with some overlap. Given the continuous, high-variance nature of ATUS data, this level of separation is consistent with expectations for behavioral data.

For the regression analysis, the baseline OLS model achieved a test R-squared value of 0.2345, while the improved model increased test R-squared to 0.2520, suggesting a modest improvement in explanatory power. All regression coefficients in the baseline model were statistically significant (p-value < 0.05), and nearly all remained significant in the expanded, improved model, demonstrating that a broad set of variables contribute to predicting leisure time. The model does not exhibit overfitting, as evidenced by the minimal difference of 0.0077 between training and test R-squared values. Furthermore, ridge regression achieved identical performance to the OLS model, confirming the coefficient estimates are stable.

To evaluate dashboard usability, we administered a short, post-interaction survey to capture usability feedback across two groups: (1) general public users with no specialized background in public health or data analysis, and (2) analytically-oriented users such as graduate students in relevant fields. Each participant was given the following task: "Please use the Leisure Predictor feature to answer the following question. How many minutes of leisure time on average does a female who is employed (at work), has no partner, has 1 child, and is Asian have? All other filters on the dashboard can remain at their default settings." They were then asked the four survey questions listed below.

1. I was able to find the information I was looking for without difficulty.
2. The filter panel was intuitive to use.
3. I understood what the two curves in the dashboard represented.
4. This tool will help me understand which population groups have the least access to leisure time.

We surveyed 6 users in the general group and 6 users in the analytical group. The user satisfaction scores in each group were averaged and reported in the table below. The target scores were achieved.

	Average User Satisfaction Score (%)	Target User Satisfaction Score (%)
General Group	90.0	80.0
Analytical Group	92.5	90.0

6. Conclusions and Discussion

Our project aimed to understand how leisure time varies across individuals in the United States and what factors shape access to leisure. Using ATUS data, we combined clustering on daily time-use patterns with a supervised regression model to capture both behavioral structure and demographic relationships.

From the regression results, we find several key demographic and socioeconomic variables to be strong predictors of leisure time. Employment status is the dominant factor, with individuals not in the labor force having substantially more leisure time (over two additional hours per day) than those actively employed. We also find a significant gender gap, with women having around 60 minutes less leisure time on average per day than comparable men. Additional factors such as number of children, education level, and work intensity (e.g., multiple jobs) show meaningful negative relationships with leisure. Each additional child is associated with roughly 17 fewer minutes of leisure on average per day, and each successive level of education is associated with a further reduction in leisure time. The overall explanatory power of the model remains moderate. While this is consistent with standard results using time-use data, it suggests a substantial portion of the variation in leisure is not captured by observable characteristics alone.

Our cluster analysis provides additional insight into this variation. The three archetypes, *Busy Workers, Overextended Individuals,* and *Time-Rich Individuals,* reflect distinct patterns in daily schedules. *Busy Workers* have the lowest leisure time (averaging 126 minutes per day), are almost entirely employed, and are more likely to be middle-aged (35-54) with a spouse present. *Time-Rich Individuals*, in contrast, are primarily older (55+) and largely not employed. They have the highest levels of leisure time, averaging

541 minutes per day, which is approximately 5.5 to 7 hours more per day than the other two groups. *Overextended Individuals* fall between these two extremes. Roughly half are employed, 62.1% are female, and 42.1% have children. *Overextended Individuals* spend more time (roughly 1-2 hours more) on household activities than the other groups. This group averages 210 minutes of leisure time per day and represents the largest share of observations (77,023), suggesting a more heterogeneous set of daily schedules than the other two archetypes.

These patterns demonstrate how overall time constraints, both from work and non-work obligations, drive differences in leisure outcomes. While these archetypes capture meaningful differences, they are not sharply distinct. The clusters show some overlap, which reflects the continuous and high-variance nature of ATUS data, where individuals often share characteristics across groups rather than fitting into rigid categories. These clusters should be interpreted as general patterns rather than strict classifications.

Taken together, the findings from the cluster and regression analyses suggest that although certain groups, such as women, individuals with children, and employed individuals, tend to have less leisure time on average, there is still substantial variation within these groups, and demographics alone do not determine access to leisure. The interactive dashboard extends these insights by allowing users to explore both patterns and predictions directly. The clustering visualizations highlight differences across schedule archetypes, while the prediction tool enables users to compare a custom profile against a baseline and view predicted leisure time and its associated uncertainty. By translating statistical outputs into an interactive interface, the dashboard makes model outputs more accessible and demonstrates how multiple factors combine to shape leisure outcomes. This approach provides a practical tool for both individuals and policymakers to better understand and visualize how different characteristics and constraints influence access to leisure.

There are several limitations to our analysis. First, the regression model only explains part of the variation in leisure time. This suggests that important unobserved (and potentially unobservable) factors influence leisure time and limit the model's ability to fully explain or predict individual outcomes. Second, ATUS data captures only a single day per respondent, providing a snapshot of each individual's time use. This may not be fully sensitive to day-to-day variability. As a result, the analysis results should be interpreted as exploratory rather than definitive.

Our findings imply that efforts to address disparities in leisure time should consider not only demographic differences, but also how constraints in daily schedules (e.g., work demands, caregiving responsibilities, job structure) shape time allocation. The persistent gender gap in leisure time further suggests that these constraints are not evenly distributed, with women facing systematically lower access to leisure even after controlling for observable factors. Policies aimed at increasing flexibility in work schedules, reducing caregiving burdens, or improving support for working families may therefore have a meaningful impact on access to leisure. Future work could extend this analysis using longitudinal data, exploring causal relationships, as well as refining both modeling approaches and the user interface.

<u>Effort Distribution:</u> All team members have contributed a similar amount of effort.

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